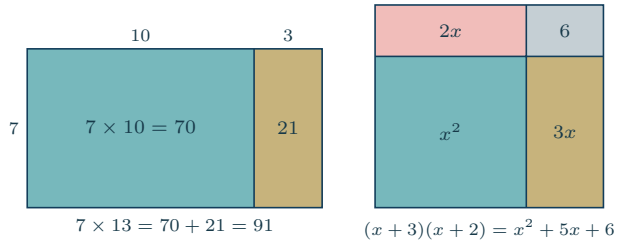
ALGEBRA LADDER 2 OF 9

Distributivity: multiplying by parts

Multiplication spreads over addition, which turns a hard product into easy pieces and an expression into a factorised form.

1 / THE RULE AND THE PICTURE

A rectangle split into two pieces has the area of both pieces together



Splitting one side of a rectangle into b and c cuts the area into ab and ac, so

$$a(b+c) = ab + ac$$

and the same picture with two split sides gives the four pieces of $(x+3)(x+2)$.

2 / MENTAL ARITHMETIC

Choose the split that makes the pieces friendly

product	split one factor	answer
17×99	$17 \times 100 - 17$	1683
6×47	$6 \times 40 + 6 \times 7$	282
8×102	$8 \times 100 + 8 \times 2$	816
13×15	$13 \times 10 + 13 \times 5$	195
15×24	$15 \times 20 + 15 \times 4$	360

The same rule works for subtraction, as in $7 \times 9 = 7 \times 10 - 7 = 63$, which is the move behind 17×99, and the rule is the reason the column method for long multiplication works at all.

3 / EXPANDING AND FACTORISING

Reading the rule from left to right expands, and from right to left it factorises

Expanding $(x+3)(x+2)$ by the area picture gives $x^2+3x+2x+6 = x^2+5x+6$, and a check with $x = 4$ gives $7 \times 6 = 42$ and $16+20+6 = 42$. Squares work the same way: $13^2 = (10+3)^2 = 100 + 60 + 9 = 169$, with the middle 60 made of two rectangles 10×3, and the tempting $100 + 9 = 109$ misses exactly those two.
Reading right to left, $6x + 9 = 3(2x + 3)$ takes out the common factor 3, and this is the move used to factorise every expression in school algebra.

4 / WHAT DOES NOT SPREAD

Multiplication spreads over addition and the reverse is false

the false rule	left side	right side
$2 + (3 \times 4)$ against $(2+3) \times (2+4)$	14	30
$12 \div (2+4)$ against $12 \div 2 + 12 \div 4$	2	9
$(3+4)^2$ against $3^2 + 4^2$	49	25
$\sqrt{9+16}$ against $\sqrt{9} + \sqrt{16}$	5	7

Powers and roots do not spread over a sum either, though they do spread over products: $(2 \times 3)^2 = 2^2 \times 3^2 = 36$. Division spreads over a sum when the sum is on top, so $(12+6) \div 3 = 12 \div 3 + 6 \div 3 = 6$, and it fails when the sum is the divisor.

5 / TRY THESE

Split, expand and check by a number

(a) Find 35×102 as $35 \times 100 + 35 \times 2 = 3570$.
(b) Expand $(x+5)(x+1)$ as $x^2 + 6x + 5$ and check with $x = 2$: $7 \times 3 = 21$ and $4 + 12 + 5 = 21$.
(c) Factorise $10y + 15$ as $5(2y + 3)$. **(d)** Show that $(a+b)^2 = a^2 + b^2$ is false by taking $a = b = 1$: the left side is 4 and the right side is 2.

ASIDE / ONE LAW, TWO DIRECTIONS

Because multiplication is commutative the law also holds with the sum on the left, so $(b+c)a = ba + ca$.

LINKS TO OTHER SHEETS

The previous sheet gives the laws used here; the matrix sheet later asks which of them survive when multiplication stops commuting.

Convention. $a(b+c) = ab + ac$ is the distributive law of multiplication over addition; squares and roots are written a^2 and $\sqrt{a}$.